121. The compound insoluble in water is

(a) Mercurous nitrate
(b) Mercuric nitrate
(c) Mercurous chloride
(d) Mercurous perchlorate

Answer: C — (c) Mercurous chloride

Chemical reason: Mercurous chloride, Hg_2Cl_2, is sparingly soluble in water; this low solubility is why it precipitates in classical qualitative analysis.

122. Which is an amphoteric oxide

(a) ZnO
(b) CaO
(c) BaO
(d) SrO

Answer: A — (a) ZnO

Chemical reason: ZnO is amphoteric: it reacts with acids to form Zn^{2+} salts and with strong bases to form zincate species. CaO, BaO, and SrO are predominantly basic oxides.

123. What is the magnetic moment of $[FeF_6]^{3-}$

(a) 5.92
(b) 5.49
(c) 2.32
(d) 4

Answer: A — (a) 5.92

Chemical reason: Magnetic behaviour depends on the number of unpaired d electrons. Writing the metal-ion d configuration for the species in 5.92 gives the required number of unpaired electrons, so it has the stated magnetic behaviour.

124. How H_2S is liberated in laboratory

(a) $FeSO_4$+H_2SO_4
(b) FeS+dil. H_2SO_4
(c) FeS+conc. H_2SO_4
(d) Elementary H^{2+} elementary S

Answer: B — (b) FeS+dil. H_2SO_4

Chemical reason: The correct choice is FeS+dil. H_2SO_4. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

125. The spin magnetic moment of cobalt in the compound $Hg[Co(SCN)_4]$ is

(a) 3
(b) 8
(c) 15
(d) 24

Answer: C — (c) sqrt(15)

Chemical reason: Magnetic behaviour depends on the number of unpaired d electrons. Writing the metal-ion d configuration for the species in sqrt(15) gives the required number of unpaired electrons, so it has the stated magnetic behaviour.

126. In which of these processes platinum is used as a catalyst

(a) Oxidation of ammonia to form HNO_3

(b) Hardening of oils

(c) Production of synthetic rubber

(d) Synthesis of methanol

Answer: A — (a) Oxidation of ammonia to form HNO_3

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability, conductivity, or corrosion resistance. That property is specifically associated with Oxidation of ammonia to form HNO_3.

127. Iron is dropped in dil. HNO_3, it gives

(a) Ferric nitrate

(b) Ferric nitrate and NO_2

(c) Ferrous nitrate and ammonium nitrate

(d) Ferrous nitrate and nitric oxide

Answer: C — (c) Ferrous nitrate and ammonium nitrate

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Ferrous nitrate and ammonium nitrate matches the relevant metallurgical process or alloy property.

128. CrO_3 dissolves in aqueous NaOH to give

(a) CrO^{42-}

(b) $Cr(OH)^{3-}$

(c) Cr_2O^{72-}

(d) $Cr(OH)_2$

Answer: A — (a) CrO^{42-}

Chemical reason: Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result CrO_4^{2-} is consistent with that behaviour.

129. KI and $CuSO_4$ solution when mixed, give

(a) $CuI_2+K_2SO_4$

(b) $Cu_2I_2+K_2SO_4$

(c) $K_2SO_4+Cu_2I_2+I_2$

(d) $K_2SO_4+CuI_2+I_2$

Answer: C — (c) $K_2SO_4+Cu_2I_2+I_2$

Chemical reason: The correct choice is $K_2SO_4+Cu_2I_2+I_2$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

130. When Cu reacts with $AgNO_3$ solution, the reaction takes place is

(a) Oxidation of Cu

(b) Reduction of Cu

(c) Oxidation of Ag

(d) Reduction of NO^{3-}

Answer: A — (a) Oxidation of Cu

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Oxidation of Cu. The choice is consistent with the expected oxidation state and stability of the product.

131. By annealing, steel

(a) Becomes soft
(b) Becomes liquid
(c) Becomes hard and brittle
(d) Is covered with a thin film of Fe_3O_4

Answer: A — (a) Becomes soft

Chemical reason: Annealing involves heating steel and then cooling it slowly. This relieves internal stress and allows a softer, more ductile microstructure to form.

132. Which of the following is more soluble in ammonia

(a) AgCl
(b) AgBr
(c) AgI
(d) None of these

Answer: A — (a) AgCl

Chemical reason: AgCl is the most soluble of the common silver halides in NH_3 because formation of $[Ag(NH_3)_2]^+$ shifts its dissolution equilibrium farthest to the right; AgBr is less soluble and AgI is nearly insoluble.

133. Potassium permagnate works as oxidising agent both in acidic and basic medium. In both state product obtained by $KMnO_4$ is respestively

(a) MnO^{2-} and Mn^{3+}
(b) Mn^{3+} and Mn^{2+}
(c) Mn^{2+} and Mn^{3+}
(d) MnO_2 and Mn^{2+}
(e) Mn^{2+} and MnO_2

Answer: D — (d) MnO_2 and Mn^{2+}

Chemical reason: The n-factor is obtained from the oxidation-state change under the stated alkaline conditions; equivalent weight is molar mass divided by that n-factor.

134. Which of the followign is the green coloured powder produced when ammonium dichromate is used in fire works

(a) Cr
(b) CrO_3
(c) Cr_2O_3
(d) $CrO(O_2)$

Answer: C — (c) Cr_2O_3

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in Cr_2O_3 has the characteristic oxidation state/ligand field that produces the observed colour.

135. Which compound does not dissolve in hot dilute HNO_3

(a) HgS
(b) CuS
(c) PbS
(d) CdS

Answer: A — (a) HgS

Chemical reason: The answer follows from solubility and complex-ion equilibria. Under the stated reagent conditions the equilibrium favours HgS, whereas the alternatives do not show the same dissolution/precipitation behaviour.

136. The least stable oxide at room temperature is

(a) ZnO
(b) CuO
(c) Sb_2O_3
(d) Ag_2O

Answer: D — (d) Ag_2O

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Ag_2O matches the relevant metallurgical process or alloy property.

137. Which of the following pari of elements cannot form an alloy

(a) Zn-Cu
(b) Fe⁻Hg
(c) Fe,C
(d) Na,Hg

Answer: B — (b) Fe⁻Hg

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Fe⁻Hg matches the relevant metallurgical process or alloy property.

138. Which of the following shows dimerisation

(a) $HgCl_2$
(b) B_2H_6
(c) $TiCl_4$
(d) SO_2

Answer: A — (a) $HgCl_2$

Chemical reason: The correct choice is $HgCl_2$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

139. Which of the following is also known as "Fools gold"

(a) Wurtzite

(b) Iron pyrites

(c) Chalcosite

(d) Silver glance

Answer: B — (b) Iron pyrites

Chemical reason: Iron pyrite, FeS_2, is called "fool's gold" because its metallic yellow appearance can resemble gold even though its composition and properties are very different.

140. Which one of the following is highest melting halide

(a) AgCl

(b) AgBr

(c) AgF

(d) AgI

Answer: C — (c) AgF

Chemical reason: The correct choice is AgF. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.